



B.E./B.Tech. DEGREE EXAMINATION, APRIL/MAY 2011
Third Semester Electrical and Electronics Engineering
EE 2204 — DATA STRUCTURES AND ALGORITHMS
(Regulation 2008) Time : Three hours Maximum : 100 marks

Answer ALL questions

PART A — (10 × 2 = 20 marks)

1. Define stack. What are the operations performed on a stack?
2. Mention the applications of list.
3. Define tree. List the tree traversal techniques.
4. Differentiate a binary tree from a binary search tree.
5. What is meant by binary heaps?
6. What is linear hashing? Specify its merits and demerits.
7. What is meant by digraph? Define the terms in-degree and out-degree with respect to a digraph.
8. Write the adjacency matrix for the following graph.
9. What is dynamic programming? Give two examples.
10. What is meant by NP-complete problem?

PART B — (5 × 16 = 80 marks)

11. (a) What is doubly linked list? Explain the algorithm in detail for inserting a node to the left and deleting a node from a doubly linked list. (16)

Or

- (b) Write an algorithm and explain how insertions and deletions are carried out in a circular queue implemented using linked list. (16)

12. (a) (i) Discuss how a node could be inserted in a binary tree? (8)
(ii) Write a procedure in C to find the Kth element in binary tree. (8)

Or

- (b) (i) Derive the expression tree for the expression $(a + b + c) + ((d * e + f) * g)$. Briefly explain the construction procedure for the above. (6)
(ii) Write routines to implement the basic binary search tree operations. (10)

13. (a) Explain with algorithm how insertion and deletion is performed in a AVL tree. Explain how the tree is balanced after the operation. (16)

Or

- (b) (i) Write a procedure in C that will traverse a linked B-tree, visiting all its entries in order of keys (smaller to larger). (10)
(ii) What is meant by collision in hashing? Explain the separate chaining collision resolution strategy. (6)



14. (a) (i) Define Graph. Briefly explain the graph traversal algorithms with an example. (8)
(ii) Find the shortest path from node 1 to 7 using shortest path algorithm. (8)

Or

- (b) With an example, explain Kruskal's algorithm for finding minimum spanning tree. (16)

15. (a) (i) Develop an algorithm for binary search using divide and conquer method. Analyze its time and space complexity. (8)

- (ii) Write a greedy algorithm to solve job sequencing with deadline problem. (8)

Or

- (b) (i) Explain any one algorithm to solve bin packing problem. (6)

- (ii) Explain backtracking algorithm design technique with an example. (10)